

Machine Learning Fundamentals

This document provides a concise introduction to the core concepts of Machine Learning (ML), including learning types, model training, evaluation metrics, and common algorithms.

1. What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence (AI) that allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed.

2. Types of Machine Learning

Supervised Learning: Learns using labeled data.
Examples: Classification and Regression.

Unsupervised Learning: Finds hidden patterns in unlabeled data.
Examples: Clustering and Dimensionality Reduction.

Reinforcement Learning: An agent learns through rewards and penalties.

3. Important Concepts

Dataset: Collection of data used for training.
Features: Input variables used for prediction.
Labels: Correct outputs in supervised learning.
Training Set: Data used to train the model.
Test Set: Data used to evaluate the model.

4. Common Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forests
- K-Nearest Neighbors (KNN)
- Support Vector Machines (SVM)
- Neural Networks

5. Model Evaluation

Common evaluation metrics include:

- Accuracy
- Precision
- Recall

- F1 Score
- Mean Squared Error (MSE)

6. Overfitting vs Underfitting

Overfitting: The model learns training data too well and performs poorly on new data.

Underfitting: The model is too simple and fails to capture important patterns.

7. Applications of Machine Learning

Machine Learning is widely used in:

- Recommendation Systems
- Fraud Detection
- Image Recognition
- Chatbots and Virtual Assistants
- Autonomous Vehicles

Quick Comparison Table

Learning Type	Main Goal	Example
Supervised	Predict outputs	Spam detection
Unsupervised	Find patterns	Customer clustering
Reinforcement	Learn through rewards	Game-playing AI

Machine Learning is one of the most impactful technologies in modern computing. Understanding its fundamentals is essential before exploring advanced fields such as Deep Learning, Natural Language Processing, and Computer Vision.